

Research Interests

Network Modeling, Traffic Simulation, Operations Research, Behavioral Psychology, Behavioral Economics, Decision Theory, Autonomous Vehicles, Transportation Equity, AI/Machine Learning, Human-Machine Interaction

EducationGeorgia Institute of Technology

Atlanta, Georgia

Doctor of Philosophy, Transportation Engineering, minor in Business Administration

- Thesis title: Carbon credit generation through voluntary travel behavior change based carbon capture
- Advisor: Dr. Srinivas Peeta, Autonomous and Connected Transportation (ACT) Lab

Master's of Science in Transportation Engineering

- Advisor: Dr. Alexander Samaylov, Aerospace Transportation and Advanced Systems (ATAS) Lab

Indian Institute of Technology (IIT) Madras

Chennai, India

Bachelor's of Technology, in Civil Engineering, Minor in Operations Research

- Bachelor's Thesis: Driving Behavior Analysis using Vehicle Trajectory Data in Mixed Traffic Conditions
- Advisor: Dr. Karthik Srinivasan

Research Experience

Jan 2023

– Present

Decision Science & Policy Lab, Georgia Tech

Graduate Research Assistant | Supervisor: Dr. Omar I. Asensio

- Conducting research on behavioral interventions and data-driven travel decision-making
- Developed methodologies for evaluating the impact of mobility choices using advanced data analytics
- Collaborated on projects integrating transportation behavior modeling and policy recommendations
- Focus Areas: Behavioral intervention design, data-driven decision-making, policy recommendations

Jan 2020

– Present

Autonomous and Connected Transportation Lab, Georgia Tech

Graduate Research Assistant | Supervisor: Dr. Srinivas Peeta

- Investigated the design & impact of positive behavioral interventions using gamification & nudge theory
- Designed and developed the TranSACT App integrating behavioral interventions & real-time feedback
- Modeled travel decisions and network effects using reinforcement learning and optimization techniques
- Assisted in writing several USDOT and NSF research proposals to secure research funding of about 4 Million dollars
- Authored conference presentations, conference proceedings, journal articles, and research reports based on over 8 USDOT, NSF, and UTC funded projects
- Led community workshops on sustainable travel at the City of Peachtree Corners
- **Focus Areas:** Sustainable travel behavior, network modeling, reinforcement learning, gamification

Aug 2017

– Dec,

2019

Aerospace Transportation and Advanced Systems Lab, Georgia Tech Research Institute

Graduate Research Assistant & Tech Intern | Supervisor: Dr. Alexander Samaylov

- Conducted research on large-scale connected and autonomous vehicles (CAV) for urban fleet management
- Developed simulation tools to optimize CAV operations and tested these on a university test track
- Contributed to NSF-funded initiatives on smart transportation for urban settings.
- **Focus Areas:** Connected and Autonomous Vehicles (CAV) and simulation-based optimization

Selected Research Projects**Testing Behavioral Interventions Using Online Surveys**

Advisors: Dr. Omar I. Asensio, Dr. Srinivas Peeta

- Designed and implemented over 50 non-priced intervention mechanisms including social norms messaging, environmental impact visualization to promote sustainable travel modes
- Conducted RCTs using Qualtrics to test intervention effectiveness on over 10,000 participants across the U.S.
- Analyzed survey data using multivariate regression and structural equation modeling to estimate behavioral shifts
- Conducted workshops with local communities to validate survey findings and refine intervention mechanisms

Field Experiment to Test Behavioral Interventions Using TranSACT Mobile App

Advisor: Dr. Srinivas Peeta

- Designing to deploy 12-week field experiments involving 300 participants in City of Peachtree Corners, GA
- Testing combinations of interventions, including real-time rewards, gamification, and health-based messaging
- Quantifying shifts in travel modes using metrics such as emissions avoided, calories burned, and air quality

Development of Transact Mobile App for Sustainable Travel Interventions

- Built the Transact app using native iOS and Android front-end development.
- Designed a Firebase and JavaScript backend for data storage and real-time intervention delivery.
- Integrated APIs for GPS tracking, user authentication, and analytics dashboards to monitor user engagement.
- Enabled scalability to support over 1,000 concurrent users with real-time messaging and travel behavior tracking.

Travel Mode Classification Using Transfer Learning

- Developed a classification model using second-by-second GPS trajectory data, leveraging pre-trained deep learning models (e.g., ResNet) for transfer learning.
- Fine-tuned the model on a dataset of 500,000 labeled GPS trajectories across walking, biking, and transit modes.
- Achieved over 85% classification accuracy using spatiotemporal feature extraction and Long Short-Term Memory (LSTM) networks.
- Automated pipeline implemented in Python with TensorFlow and scikit-learn for real-time mode detection.

Bi-Level Network Optimization for Behavioral Intervention Design

Advisor: Dr. Srinivas Peeta

- Formulated a bi-level optimization problem to design behavioral interventions in response to travel behavior.
- Implemented particle swarm optimization (PSO) to solve the high-dimensional (12,124 variables) problem.
- Demonstrated 20% reduction in emissions under optimal intervention strategies through numerical experiments.

SUMO-Based Dynamic, Personalized Intervention Design Framework

Advisor: Dr. Srinivas Peeta

- Designed a reinforcement learning framework using DDPG agent for deploying interventions dynamically in SUMO
- Integrated traffic flow, congestion, and individual user data to deliver personalized interventions every 15 minutes
- Implemented Python-based simulations for testing scenarios with varying app penetration and traffic conditions
- Achieved a 15% improvement in traffic efficiency and a 25% reduction in CO2 emissions in numerical experiments

Driving Simulator Analysis of Autonomous Vehicle Infrastructure Impacts

Advisor: Dr. Srinivas Peeta

- Investigated the impact of autonomous vehicle (AV)-specific infrastructure on human driver cognition and safety
- Conducted experiments using a high-fidelity driving simulator on the Curiosity Lab test track's digital twin
- Collected physiological and behavioral data (eye-tracking, heart rate) to evaluate cognitive load and driver response
- Analyzed human-AV interaction data using python to inform infrastructure design recommendations.

Fleet Management for Large-Scale Autonomous Vehicle Operations

Advisor: Dr. Srinivas Peeta

- Developed numerical simulations in SUMO to optimize dynamic routing and fleet management for AV operations.
- Designed an algorithm for real-time rerouting based on traffic flow and ride demand data.
- Validated the approach through experiments on the University of Tennessee, Chattanooga test track
- Achieved a 30% improvement in fleet utilization and a 20% reduction in idle vehicle time during peak hours.

Publications

Authored and co-authored peer-reviewed papers and conference presentations across TRB, INFORMS, and IEEE, advancing the fields of behavioral interventions, sustainable mobility, and intelligent transportation systems. Research spans field experiments, optimization models, and smart city data integration

Peer-reviewed journal articles

Anne, V. S. R., Liu, Y., Kibria, G., Peeta, S., & Asensio, O. I. (2025). *Promoting Sustainable Travel Modes through Health and Active Lifestyle Messaging.* PNAS Nexus. (Submitted and under peer review)

Anne, V.S.R., Liu, Y., Kibria, M.G., Peeta, S., Asensio, O.I., 2025. Replication Data for: "Promoting Sustainable Travel Modes Through Health and Active Lifestyle Messaging"; <https://doi.org/doi:10.7910/DVN/Y8HFXW>

Anand, P. A., Atmakuri, P., **Anne, V. S. R.**, Asaithambi, G., Srinivasan, K. K., Sivanandan, R., & Chilukuri, B. R. (2019). Calibration of vehicle-following model parameters using mixed traffic trajectory data. *Transportation in Developing Economies*, 5, p. 18.

Conference Papers and Presentations (Peer-reviewed)

Kibria, G., **Anne, V. S. R.**, Xu, Y., Eluru, N., & Peeta, S. (2026). "Scalable Decomposition Approach for Joint Optimization of Transit Stops and Routes to Enhance Accessibility to Points of Interest," 105th Annual Meeting of the Transportation Research Board, Washington, D.C., January 2026.

Kibria, G., Xu, Y., **Anne, V. S. R.**, Eluru, N., & Peeta, S. (2025). "Joint Optimization of Transit Stops and Routes to Improve Access to Essential Community Destinations," The 33rd ACM SIGSPATIAL, Minneapolis, MN, November 2025. Winner: **Best Paper Award**.

Anne, V. S. R., Chang, Y., & Peeta, S. (2025). Reinforcement Learning Based Framework to Design Behavioral Interventions to Improve Sustainability and Public Health. Accepted at the 2025 INFORMS Annual Meeting, Atlanta, GA, USA.

Anne, V. S. R., Liu, Y., Kibria, G., Asensio, O., & Peeta, S. (2025). "Promoting Sustainable Travel Modes through Gamified Health and Environmental Information," 104th Annual Meeting of the Transportation Research Board, Washington, D.C., January 2025. Winner: **TRB AEP35 Best Paper Award**.

Anne, V.S.R. & Peeta, S. (2024). Promoting Sustainable Mobility: Behavioral Interventions for Transportation Sustainability and Equity. Accepted in 2024 INFORMS Annual Meeting. Seattle, Washington, United States.

Xu, Y., Kibria, G., Wang, C., Tenenboim, E., **Anne, V. S. R.**, Chen, Z., & Peeta, S. (2023, December). Dynamic Visualization Platform for Travel-Related Data Integration to Support Sustainability-Based Decision-Making for Smart Cities. In 2023 IEEE 20th International Conference on Smart Communities: Improving Quality of Life using AI, Robotics and IoT (HONET) (pp. 1-6). IEEE.

Thompson, R., Stovall, J., Velasquez, D., **Anne, V.S.R.**, Samoylov, A.V. and Sartipi, M., 2018, December. All-in-One Urban Mobility Mapping Application with Optional Routing Capabilities. In 2018 IEEE International Conference on Big Data (Big Data) (pp. 1127-1133). IEEE.

Technical Reports

Anne, V. S. R., Kibria, G., & Peeta, S. (2025). "Promoting Sustainable Travel within Communities through Behavioral Interventions and Emerging Mobility Solutions," Final Project Report, Project 22, National Center for Understanding Future Travel Behavior and Demand.

Du, L., Peeta, S., Ning, Y., & **Anne, V. S. R.** (2022). Smartphone Based Incentive Framework for Dynamic Network Level Traffic Congestion Management (No. STRIDE Project H3). Southeastern Transportation Research, Innovation, Development and Education Center (STRIDE).

Anne, V.S.R., Tenenboim, E. and Peeta, S., 2022. Using Driving Simulator Environment to Determine Interactions Between User Behavior and Infrastructure Design Under Autonomous Vehicles (No. CCAT Report# 39). University of Michigan. Center for Connected and Automated Transportation. doi: <https://doi.org/10.5703/1288284317647>

Teaching:

- Served as a Teaching Assistant and developed the Transportation Networks Modeling & Analysis course (Fall 2021)
- Served as a Teaching Assistant for the Transportation Systems Analysis course (Spring 2022)
- Course mentor for undergrad Vertically Integrated Projects course since Fall '20 & Grand Challenges course since Fall '21

Skills

Programming Languages: Python, R, Swift, Kotlin, Java, C++, Bash

Frameworks & Libraries: PyTorch, HuggingFace, DeepSpeed, TensorFlow, Keras, WandB, Matplotlib

Tools: SUMO, Carla, Git, Docker, Jetpack Compose, SwiftUI, Streamlit.io